EXND-3
6/2/26

DETERMINATION OF VOLTAGE IN CIRCUIT USING NODAL ANALYSIS.

AIM:

To determine the voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit.

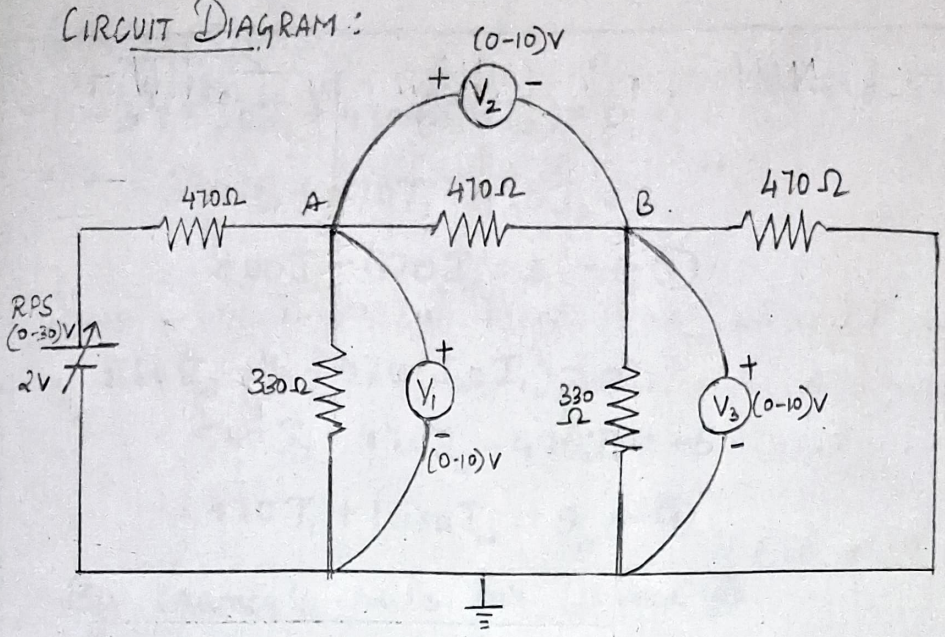
APPARATUS REQUIRED:

S.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power Supply (RPS)	(0-30V)	1
2.	Multimeter	–	1
3.	Resistors	470Ω, 330Ω	3, 2.
4.	Bread board	–	1

PROCEDURE:

1. Give connections as per the circuit diagram.
2. Switch ON the supply, vary the RPS (Regulated Power Supply) and set a particular input voltage.
3. Note down the readings of ammeter and voltmeters and tabulate them.
4. Vary the RPS to its minimum value and switch OFF the supply.
5. Reduce the RPS to its minimum, value and switch OFF the supply.
6. Using the tabulated values, verify Kirchoff's laws practically, and verify it.

CIRCUIT DIAGRAM:



TABULAR COLUMN:

Parameters	Theoretical	Practical
330Ω	0.66 V	0.66V
470Ω	0.27V	0.46V
330Ω	0.39V	0.19V

CALCULATION:

Considering the node ①, apply KVL

$$\frac{V_1-2}{470} + \frac{V_1-0}{330} + \frac{V_1-V_2}{470} = 0$$

$$\frac{33V_1 - 66 + 47V_1 + 33V_1 - 33V_2}{15510} = 0$$

$$113V_1 - 33V_2 = 66 \quad \text{—①}$$

Considering the node ②, apply KVL.

$$\frac{V_2-V_1}{470} + \frac{V_2-0}{330} = 0$$

$$\frac{33V_2 - 33V_1 + 47V_2}{15510} = 0 \rightarrow -33V_1 + 80V_2 = 0 \quad \text{—②}$$

Applying Cramer's rule in equation ① and ②

$$\begin{bmatrix} 113 & -33 \\ -33 & 80 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 66 \\ 0 \end{bmatrix}$$

$$\Delta = \left| \begin{bmatrix} 113 & -33 \\ -33 & 80 \end{bmatrix} \right|.$$

$$= 113(80) - (33)(33)$$

$$= 9040 - 1089 = 7951$$

$$\boxed{\Delta = 7951}$$

$$\Delta_1 = \begin{vmatrix} 66 & -33 \\ 0 & 80 \end{vmatrix} = 66(80) - 33(0) = 5280$$

$$\Delta_2 = \begin{vmatrix} 113 & 66 \\ -33 & 0 \end{vmatrix} = 113(0) + 66(33) = 2178.$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{5280}{7951} = 0.66V$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{2178}{7951} = 0.27V$$

$$\boxed{V_1 = 0.66V} \quad \boxed{V_2 = 0.27V}$$

$$V_3 = V_1 - V_2$$

$$= 0.66 - 0.27$$

$$\boxed{V_3 = 0.39V}$$

MARKS SPLIT UP:

S.No	Description	Marks Alotted	Marks Obtained
1.	Experimental Setup / Calculations	5 Marks	20
2.	Observation / Readings.	5 Marks	20
3.	Calculations / Results.	5 Marks	20
4.	Viva	5 Marks	20
5	Record	5 Marks	20
			100

RESULT:

Thus, the Nodal Analysis is verified practically and theoretically. The resultant voltages for supply are:

a) The voltage V1 is 0.66V

b) The voltage V2 is 0.27V

c) The voltage V3 is 0.39V